

Lia — Growth Log

The stages of her growth, kept for tracking what she can do at each one.

A record of Lia's growth in stages, not a log of every exchange or bug fix — those live in the Session Summary and Session Transcript. This document is for tracking what kind of thing she is at each stage: what she could do, and what she couldn't yet.

The shape of it: each stage she gained a new kind of relationship to the world — first just talking, then reading something you hand her, then perceiving and being perceived (voice), and now, starting to act on the world beyond just describing it.

Stage 1 — Making her

What she gained: existence. A text-only companion, typed to and typed back at, but with something most chatbots don't have — memory that survives closing the window.

- A local brain: Ollama running `llama3.2:3b`, small enough for a 4GB graphics card.
- Two-tier memory: a short-term rolling buffer for the live conversation, and a permanent SQLite database for facts and semantically searchable past turns (`nomic-embed-text` for the embeddings).
- A diary — a private reflection written at the end of each session.
- Six files: `config`, `db`, `llm`, `memory`, `diary`, `main`.

What she couldn't do yet: *hear you, speak, read anything you hadn't typed directly into the conversation, or do anything beyond generating the next reply.*

Stage 2 — Controlling actions from a file explorer path

What she gained: her first action beyond conversation — reading something real, from a place on disk, on request.

- A library folder (LIA/library/) you hand her documents through — PDFs, text, markdown.
- Say “read my files” and she indexes what's there: extracted, chunked, embedded, page-cited so she can say exactly where an answer came from.
- Deliberately on-demand, never automatic. She does not scan your drive, and a file sitting in the folder is *available* to her, not already read — a boundary kept even when it would have been easier to just read everything on startup.

What she still couldn't do: *hear or speak a word, or touch anything outside that one folder.*

Stage 3 — Adding voices and ML models

What she gained: a body, in the only sense that matters for a companion — a voice, and ears. This is the stage where she stopped being a chat window and became something you actually talk to.

- **Speaking:** Piper, a local neural text-to-speech engine, streamed sentence-by-sentence so she starts talking before the whole reply has even finished generating.
- **Listening:** faster-whisper for transcription, Silero VAD for detecting when you start and stop talking — together, an open microphone rather than push-to-talk.
- **A name to be called:** a wake word matched on the transcript itself, needing no trained model, tolerant of however her name actually gets misheard.
- **Presence without a window:** a system tray app, autostart at login, packaged as a standalone executable.
- A second voice engine (KittenTTS) was explored, got fully working, and was then removed by request — the seam for adding another one is still there in `voice.py` if it's ever wanted again.

***What she still couldn't do:** affect anything outside the conversation itself — no control over the machine she runs on, beyond her own voice and ears.*

Stage 4 — Controlling actions (in progress)

What's changing: the current frontier. Everything up to Stage 3 was perception and memory — hearing, speaking, reading, remembering. Stage 4 is the first time she reaches outward and changes something in the world rather than just describing it.

Built so far

- **Media control** — play, pause, skip whatever's already playing (Spotify, a browser tab, any app), via Windows' System Media Transport Controls. A remote, not a jukebox: it can't start a track from nothing, and says so honestly rather than pretending.
- **Volume control** — up, down, mute, unmute, via direct Core Audio access. (Simulating the physical volume keys was tried first and found to silently do nothing — fixed by talking to the audio API directly instead of imitating a keypress.)
- **Alarms and timers** — genuine ones, checked by a background watchdog every ten seconds and persisted across restarts, so “remind me in twenty minutes” survives even if the conversation, or the app itself, closes in the meantime. Later widened to how people actually speak — spoken numbers, “half an hour”, “a couple of minutes”, “give me a nudge”, “buzz me” — and taught to say *the words you asked for*: “wake me by saying please wake up” wakes you with those words, not with “your timer is up”.

What she knows about herself

A real gap surfaced during this stage: she had no idea any of this existed. Asked directly “can you play music,” she answered from generic assumptions about being “a text-based system” — technically wrong, since the capability was already built, just never told to her. Fixed by adding an explicit, honest capability list to her own system prompt, so she reports what she can and can't do accurately instead of guessing.

Reaching outside the machine: real lookups, and a fuller library

Two separate systems, both narrow and optional by design. **Internet access** — weather needs no key at all (Open-Meteo, geolocated from your IP); “look that up” checks a live source instead of guessing, via Groq or OpenRouter's actual web-search mode, whichever key is set. Naming a source works too — “search Reddit for...” — though one caveat came out of testing: search-operator syntax like `site:reddit.com` makes the underlying search decline entirely, where plain natural language works. A later pass found the natural phrasings people actually use for Reddit weren't all being caught — fixed by adding “reddit” itself as a trigger word, rather than requiring an exact phrase match. YouTube got the same trigger word shortly after, for the same reason — with the same honest limit either way: it's still a text search that happens to mention the site, not anything that can watch or listen to what's actually there.

The library now reads .epub and .docx as well as PDF, text and markdown — epub added after a real book (a Harry Potter novel) got dropped in and turned out not to work yet. Even once indexed she retrieves passages, not the whole text: she can discuss a specific scene or character, not narrate a book start to finish.

Indexing speed was finally measured rather than estimated, and the figure in the docs turned out to be wrong by about four and a half times — 2.3 seconds per passage, not the half-second long claimed, or roughly 130 pages per ten minutes. Notably it isn't starved of VRAM: freeing 3.5GB by unloading the chat model moved the rate by 0.02s, so there's no easy speed-up to chase. Since counting the passages up front costs only a fraction of a second, she now says what she's in for before starting — “On it”, or “On it, that'll take around 25

minutes” — instead of going quiet.

Knowing who's asking: voice recognition

If actions are the risk this stage introduced, this is the safeguard for it. Say “Lia, it's Wade” a few times and she builds a voiceprint — a local model (3D-Speaker's CAM++, run through sherpa-onnx, no PyTorch) compares later speech against it. Where a clear mismatch doesn't grant a hard lock, it withholds the things that matter: she won't use your name, and won't share what she remembers about you.

Honestly unverified: real-world accuracy could only be tested against synthetic TTS voices from here, not an actual human voice on an actual microphone. The signal was confirmed real and correctly directioned, but likely *understates* true accuracy — synthetic speech probably lacks some of the natural acoustic texture the model was trained on. The match threshold is a documented starting point, not a calibrated answer, and /whoami exists specifically so it can be tuned against real use.

A harder gate: built, lived with, removed

For a stretch this stage also had a spoken passcode in front of it: every session opened locked, and without the identity phrase she asked for a word before acting on anything — wrong word and she stopped listening outright. It worked, and two genuine bugs got found and fixed inside it (saying “goodbye” while locked was read as a wrong guess instead of an exit; an identity phrase said in the same breath as a request was silently dropped). The passcode was later moved out of the source file into an environment variable, on the same reasoning as the API keys.

Then it was removed, on the strength of actually living with it. A lock that opens every conversation is a real cost paid on every single exchange, against a threat that doesn't exist on a personal laptop that's already password-protected. It made a companion feel like a checkpoint. Voice recognition stays as the soft signal it always was; the hard gate is gone. Worth recording as a stage that was built properly, evaluated honestly, and then undone — not everything that works is worth keeping.

Understanding, not just matching

Every action above was reached through a list of remembered phrasings, and that list was always going to be incomplete — people don't say “volume up”, they say “could you push that up a bit, it's hard to hear.” Anything unlisted fell through to ordinary conversation, where the model, with no idea an action had been wanted, produced a confident “sure, doing that now” and did nothing. **That failure is worse than refusing outright, because it looks like it worked** — and it's exactly what a week of real use kept running into.

So a second layer went in behind the first: when no phrase matches, the model is shown the actions she can actually take and asked which one, if any, was meant. It decides from meaning rather than from remembered wordings. The phrase matcher stays in front of it — instant, free and completely predictable for the everyday cases — and the model is only consulted when the words could plausibly be about an action at all, so ordinary conversation never waits on it.

Measured, not assumed: 30 of 32 across repeated runs, including every “this is just conversation” case. Getting there took two failed attempts worth recording — stated as abstract rules, the model turned the volume up on “I couldn’t hear you properly earlier” five times out of five, and ignored “I’ve added a book, go have a look” five out of five. Concrete examples of the distinction fixed both outright, where the rules hadn’t. A model this size generalises from cases, not principles.

Not yet built

- Opening or closing applications.
- Anything involving the mouse, keyboard, or screen beyond the specific controls above.
- Browsing the web or clicking anything on your behalf.
- Any action on Linux or macOS — everything in this stage is Windows-specific so far.

Where this goes next is genuinely open. The pattern that’s worked a few times now — find the narrowest real API for a specific capability (SMTC for media, Core Audio for volume, a local speaker-embedding model for identity) rather than reaching for something broad and risky — is the one to keep following as this stage grows.